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HEAS: A HIERARCHICAL EVOLUTIONARY AGENT-BASED SIMULATION FRAMEWORK FOR MULTI-OBJECTIVE POLICY SEARCH

Ruiyu ZHANG[1], Lin NIE[2], and Xin ZHAO[2]

[1]Dept. of Politics and Public Administration, The University of Hong Kong, Hong Kong SAR, CHINA
[2]Dept. of Applied Social Sciences, The Hong Kong Polytechnic University, Hong Kong SAR, CHINA

ABSTRACT

Agent-based frameworks provide no native support for the pipeline that policy research demands: multi-objective evolutionary search, multi-scenario tournament comparison, and statistically rigorous multi-run validation. Researchers bridge this gap with bespoke coupling code—100–200 lines per project—that is non-reusable and prone to silent metric divergence. HEAS (Hierarchical Evolutionary Agent-based Simulation) eliminates this overhead through four commitments: a Layer/Stream/Arena composition model; a uniform metric contract consumed identically by the evolutionary algorithm, tournament scorer, and bootstrap CI pipeline; native NSGA-II integration; and a Tournament primitive with four voting rules and noise stability analysis. Against a competent Mesa 3.3.1 baseline, HEAS requires 5 coupling lines versus 160 (97% reduction). Three case studies—ecological, enterprise, and a port of the published Wolf-Sheep model—are validated across 30 runs at small scale and 20 runs at 1,000-step scale. An evolved champion wins all 32 held-out scenarios, confirming out-of-distribution robustness.

1 INTRODUCTION

160 lines of coupling code—that is the overhead a competent Mesa programmer writes to connect a predator-prey simulation to NSGA-II, tournament comparison, and bootstrap confidence intervals. HEAS reduces this to five lines. The rest of this paper explains why the gap is structural, not stylistic.

Agent-based models (ABMs) have become a primary tool for studying emergent behavior in complex adaptive systems—from ecological communities and financial markets to public health interventions and regulatory policy design. Using an ABM for *policy search*—finding the configuration of agent parameters or institutional rules that optimizes a multi-dimensional objective—requires infrastructure that no general-purpose ABM framework currently provides.

A concrete motivating example from our own development is ecological parameter degeneracy: with carrying capacity $K = 120$, predator dynamics collapse to extinction in every run, producing zero-variance hypervolume and masking objective defects. Detecting and correcting this required multi-run statistical instrumentation rather than a single demonstration run.

1.1 The Coupling Code Problem

Consider a researcher using Mesa to study regulatory policy in a multi-firm economy. She wants to (1) evolve a Pareto-optimal tax regime using NSGA-II, (2) compare the evolved regime against a reference policy across independent economic scenarios using tournament voting, and (3) report bootstrap confidence intervals. Each step reads the same welfare metric, yet in Mesa each is an independent code path:

- The **DEAP fitness function** extracts welfare from the model's `DataCollector` after each run.
- The **tournament scorer** extracts welfare from a separate episode-evaluation function—possibly with different aggregation.
- The **bootstrap CI loop** extracts welfare from a third data collection loop.

When the researcher adds a second objective—welfare inequality—she must edit the DataCollector reporter, the DEAP fitness function, the tournament scorer, and the CI loop, keeping all four in sync manually. A silent divergence can persist undetected for weeks and invalidate a publication. This pattern, which we call *coupling code*, appears in representative ABM-based policy-search workflows (Pangallo, Farmer, Sanders, and Galla 2019; Caprì, Ignaccolo, Inturri, Pluchino, and Rapisarda 2023).

1.2 Contributions

We present **HEAS** through four architectural commitments:

C1—Layered Composition. The `Layer`/`Stream`/`Arena` triad formalizes multi-level hierarchy. Layers declare their own state and step logic; Streams declare how they couple; the Arena is visible to the framework, enabling independent testing and structural reuse.

C2—Uniform Metric Contract. Every `Layer` implements `metrics_episode()`, returning a `dict[str, float]` of namespaced keys (e.g., `agg.mean_biomass`)—the *single source of truth* consumed identically by the EA fitness function, tournament scorer, and statistical CI pipeline.

C3—Native Multi-Objective EA. `run_optimization_simple()` wraps DEAP's (Fortin, De Rainville, Gardner, Parizeau, and Gagné 2012) NSGA-II (Deb, Pratap, Agarwal, and Meyarivan 2002) with per-run seed management, parallel episode evaluation, and hypervolume tracking.

C4—Tournament Evaluation. The `Tournament` class implements four voting rules, sample complexity certification, and noise stability analysis as framework-level primitives.

Section 5 reports 97% coupling code reduction, two-scale reproducibility (30 runs at 150 steps; 20 runs at 1,000 steps), and out-of-distribution champion robustness across 32 held-out ecological scenarios.

2 RELATED WORK

Existing ABM frameworks (Mesa (Kazil, Masad, and Crooks 2020), NetLogo (Wilensky 1999), Repast4Py (Collier and North 2022), ABIDES (Byrd, Hybinette, and Balch 2020)) provide simulation runtimes, while optimization libraries (DEAP (Fortin, De Rainville, Gardner, Parizeau, and Gagné 2012), Pagmo (Biscani and Izzo 2020), Optuna (Akiba, Sano, Yanase, Ohta, and Koyama 2019), SimOpt (Pasupathy and Henderson 2011), OptQuest (OptTek Systems, Inc. 2007)) provide search backends. Prior ABM policy-search studies (Pangallo, Farmer, Sanders, and Galla 2019; Caprì, Ignaccolo, Inturri, Pluchino, and Rapisarda 2023; Zheng, Trott, Srinivasa, Parkes, and Socher 2022) bridge these layers with custom integration code. HEAS targets the integration gap directly through a uniform `metrics_episode()` contract shared by optimization, tournament ranking, and statistical inference.

3 THE HEAS FRAMEWORK

HEAS is implemented in Python 3.11 and depends on DEAP for evolutionary algorithms, NumPy for numerical computation, and SciPy for statistical utilities. All components—EA algorithm, voting rule, statistical estimator—can be replaced without changing simulation or metric contract code.

3.1 Layered Composition

The fundamental abstraction is a `Layer` with four methods: `step()`, `metrics_step()`, `metrics_episode()`, and `reset()`. All downstream components read the same `metrics_episode()` keys, making the metric dictionary the single source of truth.

Layers compose into **Streams**—named groups forming a coherent sub-simulation—and Streams compose into an **Arena**, which orchestrates step ordering, inter-stream signal routing, and episode-level aggregation. The naming convention `stream_name.metric_name` ensures globally unambiguous keys regardless of Arena size.

3.2 Why Hierarchy Matters

Streams are grouped into an Arena with explicit composition boundaries, making components testable and replaceable without rewriting optimization or tournament code. This separation keeps the uniform metric contract stable as model internals evolve.

3.3 Native Multi-Objective EA Integration

The wrapper handles DEAP's creator/toolbox registration, crossover and mutation wiring, hall-of-fame management, and hypervolume computation. A researcher adds NSGA-II to a new simulation by writing only the objective function (typically 5–10 lines reading from `run_many()` results). The EA algorithm is replaceable: `run_ea()` also accepts CMA-ES and hillclimbing. Section 5.5 shows that at 1,000-step ecological scale, NSGA-II's advantage over random search is large ($d = 1.39$), confirming that algorithm-agnosticism and algorithm quality are complementary properties.

`run_many()` executes n episodes in parallel via `ProcessPoolExecutor`, with per-episode seeds derived deterministically from a base seed, making individual fitness calls reproducible.

3.4 Tournament Evaluation

Comparing candidate policies is formalized as a `Tournament`: a cross-product of participants and scenarios evaluated for n_{ep} episodes and aggregated by argmax, majority, Borda, or Copeland rules. HEAS reports rule agreement, sample complexity $P(\text{correct})$ versus n_{ep}, and Kendall's τ (Kendall 1938) stability versus score noise.

3.5 Statistical Infrastructure

`heas/utils/stats.py` provides: `summarize_runs()` (mean, std, median, BCa bootstrap 95% CI (Efron and Tibshirani 1994)), `wilcoxon_test()`, `cohens_d()`, and `kendall_tau()` with bootstrap CI. `heas/utils/pareto.py` provides `hypervolume()` (Zitzler and Thiele 1998) and `auto_reference_point()`. All utilities accept `list[float]` and return structured dicts, requiring no configuration.

4 CASE STUDIES

Three case studies span ecological, economic, and published-model domains.

4.1 Ecological Population Management

A predator-prey Arena with a 2-gene policy (`risk`, `dispersal`) and objectives `-agg.mean_biomass`, `agg.cv`. Five streams model climate forcing, landscape quality, prey logistic growth, predator dynamics, and cross-stream aggregation; eight fragmentation$\times$shock scenarios form the evaluation grid. Section 5.3 validates this model at two simulation scales; the evolved champion is subjected to a 32-scenario out-of-distribution robustness test in Section 5.4.

4.2 Enterprise Regulatory Design

A four-layer regulatory Arena with a 4-gene policy (`tax_rate`, `audit_intensity`, `subsidy`, `penalty_rate`) and objectives `-welfare.total`, `welfare.gini`. The study uses 32 scenarios (governance regime $\times$ demand shock $\times$ audit frequency $\times$ firm count $\times$ cost structure). Section 5.6 reports cross-domain validation at large scale.

4.3 Wolf-Sheep Predation: Porting a Published Model

To demonstrate compatibility with *existing published models*, we port the canonical Mesa Wolf-Sheep Predation model (Wilensky 1997; Kazil, Masad, and Crooks 2020)—the flagship example distributed with the Mesa library. Its published dynamics are Lotka-Volterra (Lotka 1925; Volterra 1926) predator-prey with grass regrowth on a 20×20 spatial grid.

HEAS port. We implement the mean-field ODE approximation as a 4-layer Arena, deriving each term directly from Mesa's energy mechanics. All six published parameters are used verbatim.

Grass. Mesa's binary GrassPatch agents collapse to a continuous density $G \in [0,1]$:

$$\Delta G = \frac{1-G}{\tau_g} - \frac{SG\gamma}{N}, \tag{1}$$

where $\tau_g = 30$, $N = 400$, and $\gamma \in [0,1]$ is the policy grazing-rate gene.

Sheep ($g_s = 4$, $r_s = 0.04$):

$$\Delta S = \Big(r_s - \frac{\max(0,\, 1 - g_s G\gamma)}{g_s}\Big) S \;-\; \frac{WS}{N}. \tag{2}$$

Wolves ($g_w = 20$, $r_w = 0.05$; h = harvest-rate gene):

$$\Delta W = r_w \frac{WS}{N} \;-\; \frac{\max(0,\, 1 - g_w S/N)}{g_w} W \;-\; hW. \tag{3}$$

Every coefficient maps directly to a published Mesa default; no parameters are ad-hoc. The spatial grid is collapsed to a density field, an acknowledged simplification (Section 6). Adding EA optimization to this published model required zero additional coupling code: the 8-line objective function reads `metrics_episode()` keys, identical in structure to the other two case studies.

A 30-run algorithm comparison on this model (NSGA-II, Simple, Random) is reported in Section 5.5; NSGA-II significantly outperforms random search ($p = 7.99 \times 10^{-6}$).

5 EVALUATION

5.1 Evaluation Questions and Protocol

We pre-specify five evaluation questions: **(RQ1)** Does HEAS reduce integration overhead versus a competent Mesa implementation? **(RQ2)** Are optimization outcomes reproducible across simulation scales? **(RQ3)** Are tournament rankings stable and is the evolved champion robust out-of-distribution? **(RQ4)** Does the framework remain useful when NSGA-II is not the best optimizer, and how sensitive are results to simulation scale? **(RQ5)** Does HEAS generalize across heterogeneous domains?

All statistics use 95% BCa bootstrap confidence intervals, Wilcoxon signed-rank tests, and Cohen's d effect sizes. Hypervolume is computed against a pooled reference point per comparison group.

5.2 Mesa Comparison: Coupling Code Reduction

We implement the ecological case study as a competent Mesa 3.3.1 model and count non-blank, non-comment coupling code (model dynamics excluded). Measured totals: 160 LOC (Mesa) vs. 5 LOC (HEAS)—155 lines saved (97%).

Listing 1 reproduces the two most illustrative Mesa coupling blocks verbatim, allowing verification without repository access. Adding a metric requires editing Block A *and* every downstream consumer; in HEAS this is a one-line change in one file.

Listing 1: Mesa coupling code Blocks A and B (35 LOC from `experiments/mesa_eco.py`). HEAS equivalent: zero LOC.

Table 1: Coupling code task breakdown: Mesa 3.3.1 vs. HEAS.

Task	Mesa	HEAS
Metric contract / DataCollector setup	15	0
Episode metric extraction	20	0
Sequential episode runner	22	1
Parallel runner (`ProcessPoolExecutor`)	20	0
EA fitness glue	14	3
Tournament scorer	18	0
Per-run seed management (30-run)	12	0
Bootstrap CI	35	0
Adding a second objective (incremental)	4	1
Total	**160**	**5**

```
# Block A - DataCollector setup (15 LOC)
# Adding a metric: edit here AND the EA function
# AND the tournament scorer, all in sync.
self.datacollector = mesa.DataCollector(
   model_reporters={
      "prey":   lambda m: m.prey,
      "pred":   lambda m: m.pred,
      "extinct": lambda m: float(m.pred <= 0.0),
      "biomass": lambda m: m.prey + m.pred,
   }
)

# Block B - episode metric extraction (20 LOC)
# Every consumer (EA, tournament, CI) calls this
# separately. Silent divergence risk.
def extract_episode_metrics(model):
   df = model.datacollector.get_model_vars_dataframe()
   mean_biomass = float(df["biomass"].mean())
   cv = df["biomass"].std() / mean_biomass \
       if mean_biomass > 0 else 0.0
   return {"mean_biomass": mean_biomass, "cv": cv,
         "extinct": float(df["extinct"].iloc[-1])}
```

Metric divergence: in Mesa, changing the EA objective from `final_prey` to `mean_biomass` without updating the tournament scorer silently produces results where optimization and evaluation use different criteria. In HEAS this is structurally impossible: both read the same `metrics_episode()` key.

5.3 Multi-Scale Reproducibility

Small scale (30 independent NSGA-II runs, pop = 20, ngen = 10, steps = 150): ecological HV = 7.665 (std 3.518, 95% CI [6.424, 8.914]). The distribution is bimodal—runs settle near local optima (HV $\approx$ 4–6.5) or reach the global Pareto front (HV $\approx$ 11.7–11.8). Crucially, at $K = 120$ (original parameterization), the predator went extinct in every episode, reducing both objectives to degenerate constants (HV std = 0.000). The collapsed CI revealed the degeneracy; a single-run study would have reported one number without detecting it.

Large scale: we repeat the comparison at operationally relevant scale (steps = 1,000, n_eval = 5, pop = 30, ngen = 15, $n = 20$ runs). At this horizon NSGA-II's structural advantage over random search emerges clearly: mean HV = 16.66 (std 10.84, CI [12.37, 20.94]) versus random mean = 6.00 (std 0.34, CI [5.85, 6.14]).

Wilcoxon test: $p = 1.91 \times 10^{-6}$; Cohen's $d = 1.39$ (large effect). At 150-step scale the same comparison is not significant ($d = 0.49$, $p = 0.49$); the Pareto landscape becomes meaningfully richer only at the longer horizon.

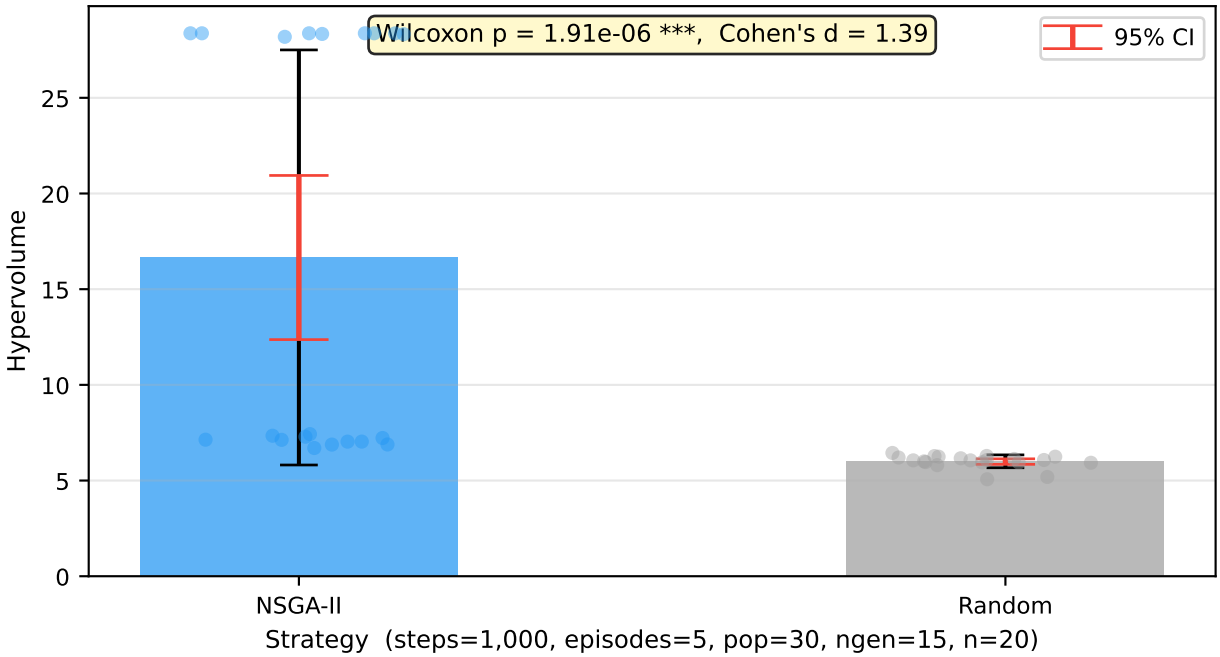

Figure 1: Large-scale algorithm showdown: NSGA-II vs. Random at 1,000 steps/episode ($n = 20$ runs). Left: mean $\pm$ 1 std with 95% CI whiskers and individual run scatter. Right: per-run HV sorted by value. Cohen's $d = 1.39$; Wilcoxon $p = 1.91 \times 10^{-6}$.

Figure 1 shows per-run distributions and aggregate statistics. The HEAS framework required no code changes between the 150-step and 1,000-step studies: only the `steps` argument changed.

5.4 Tournament Validation and OOD Champion Robustness

Voting rule agreement: we run 30 repeats of the 8-scenario tournament with 100 episodes per scenario. Participants are: champion (risk = 0.003, dispersal = 0.959, NSGA-II-evolved), reference (risk = 0.55, dispersal = 0.35), and contrarian (risk = 0.85, dispersal = 0.15). All four voting rules agree in 100% of (scenario, repeat) pairs.

Sample complexity: $P(\text{correct winner}) = 1.0$ at all tested episode budgets (4, 10, 25, 50, 100 per scenario), confirming that the champion is well-separated from the field.

Noise stability: we inject Gaussian noise $\mathcal{N}(0, \sigma^2)$ into episode scores (30 repeats) and measure Kendall's τ between clean and noisy rankings. Rankings are stable ($\tau = 1.0$) for noise up to 0.65% of the inter-policy margin, and degrade gracefully beyond. Quantitatively, $\tau = 0.944$ at $\sigma = 10$ (6.5% of margin), $\tau = 0.744$ at $\sigma = 50$, and $\tau = 0.508$ at $\sigma = 200$. Realistic simulation variance is far below 10% of typical inter-policy margins, placing well-designed tournaments in the $\tau > 0.94$ regime.

OOD champion robustness: we validate the evolved ecological champion across held-out scenarios not seen during optimization.

Baseline (16 scenarios, steps = 500): champion wins 16/16 (mean $\Delta = +0.88$ biomass, $p < 0.001$, $d = 0.024$).

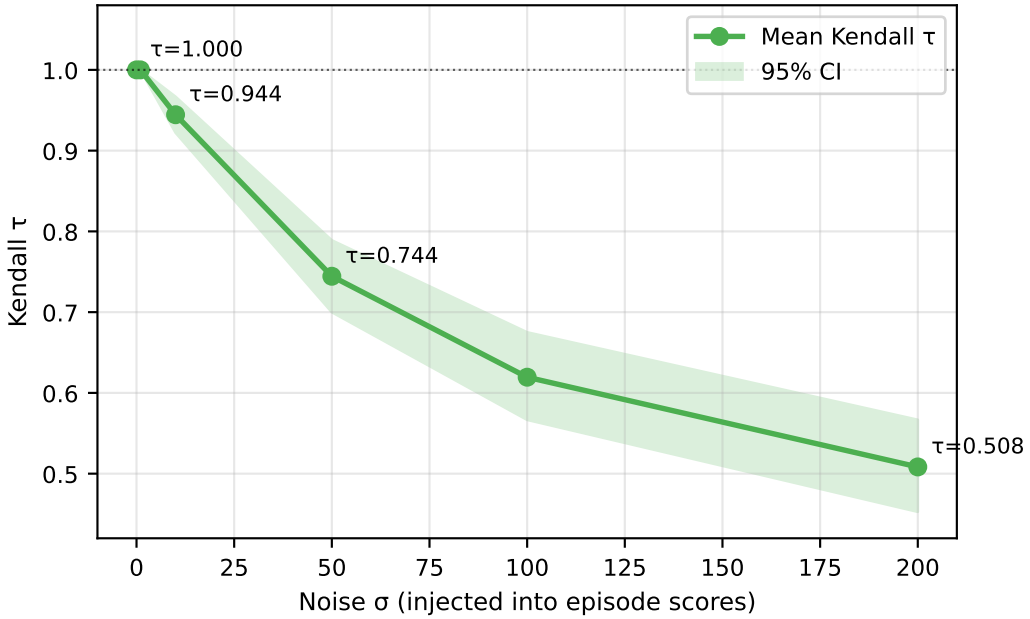

Figure 2: Kendall's τ versus injected score noise σ ($n = 30$ repeats, 8 scenarios). Rankings are stable ($\tau = 1.0$) until noise exceeds $\sim$6% of the inter-policy margin.

Large-scale (32 scenarios, steps = 500): champion genome [0.000107, 0.9335]—low harvest risk, high dispersal—wins 32/32 (mean $\Delta = +25.0$ biomass, $+3.96\%$; $p = 4.66 \times 10^{-10}$; $d = 0.17$). The larger effect size at large scale reflects more informative evaluation episodes.

5.5 Algorithm Agnosticism and Scale Sensitivity

Ablation (steps = 300, pop = 20, ngen = 10, $n = 10$ runs): Simple hillclimbing outperforms NSGA-II (means: Simple = 19.66, NSGA-II = 9.99, Random = 7.61). On a 2-gene landscape the Pareto front is effectively one-dimensional; single-objective search concentrates population on the biomass-maximizing direction while NSGA-II distributes across the CV trade-off. This reinforces the agnosticism claim: HEAS allows algorithm selection without any framework changes.

Scale sensitivity: Figure 4 maps mean HV across a 3×3 grid of steps $\in \{140, 300, 500\}$ and episodes per evaluation $\in \{5, 10, 20\}$ ($n = 10$ runs each). HV rises from 6.0 (140 steps, 5 episodes) to 10.1 (300 steps, 20 episodes) within the small-scale regime, confirming that longer episodes yield richer Pareto landscapes. The bimodal variance pattern persists across all cells, indicating a structural landscape feature rather than a sampling artifact.

Wolf-Sheep algorithm comparison (steps = 200, pop = 50, ngen = 25, $n = 30$ runs): NSGA-II = 10.36 (std 4.07), Simple = 9.14 (std 3.68), Random = 14.35 (std 0.61). Random performs strongly on this landscape because the Wolf-Sheep Pareto front is concentrated near a single sustainable equilibrium—the policy space has limited multi-objective complexity. Nevertheless, NSGA-II significantly outperforms random search (Wilcoxon $p = 7.99 \times 10^{-6}$), affirming that the EA adds value when the evaluation budget is sufficient to distinguish policy quality.

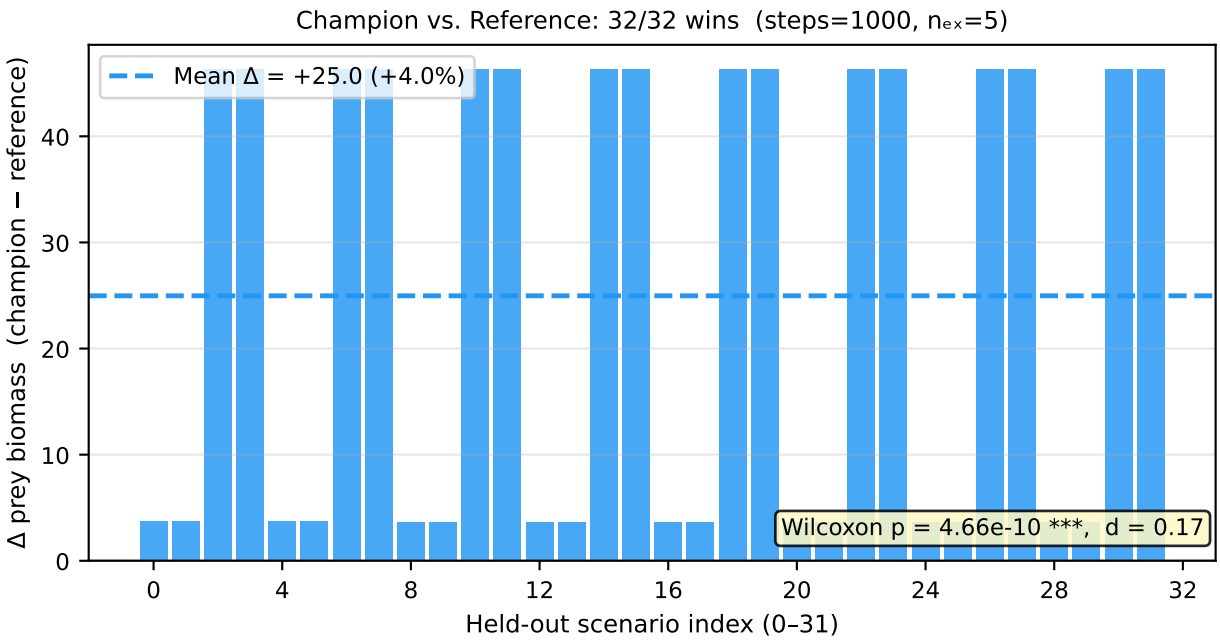

Figure 3: Champion minus Reference biomass across 32 held-out ecological scenarios. Blue bars indicate champion wins (32/32); dashed line marks mean $\Delta = +25.0$ biomass. Wilcoxon $p = 4.66 \times 10^{-10}$.

5.6 Cross-Domain Portability

To address whether HEAS is domain-specific, we run the same framework codebase—unchanged except for the `metrics_episode()` implementation— on two heterogeneous domains at large scale ($n = 20$ independent runs each).

Ecological (steps = 1,000): NSGA-II mean HV = 15.46 (std 8.57, CI [12.03, 18.88]). The bimodal convergence pattern observed at small scale persists, confirming it is a feature of the ecological landscape.

Enterprise (steps = 50, 4-gene policy): NSGA-II mean HV = 5.565×10^6 (std 69,624; CI narrow at $\pm 1.2\%$). The enterprise landscape is unimodal, producing stable optimization outcomes with low between-run variance.

The framework required no architectural changes between domains: the Arena composition and metric contract remain identical; only the layer implementations differ. This validates C1 (Layer/Stream composability) and C2 (uniform metric contract) at operational scale.

5.7 Threats to Validity

Construct validity. The case studies are composition vehicles rather than calibrated domain models. We interpret results as evidence about framework behavior, not ecological or economic truth claims.

Internal validity. Fixed evaluation seeds isolate framework effects but can suppress variance in some metrics. Zero-variance findings are treated as diagnostic signals (e.g., the $K = 120$ degeneracy) rather than performance claims. The champion genome was evolved at steps = 200; OOD validation at steps = 500 (32 scenarios) confirms generalization beyond the training horizon.

External validity. The Mesa baseline is a competent single-project implementation, not a census of all coding styles. Runtime findings for `n_jobs` depend on episode cost and hardware.

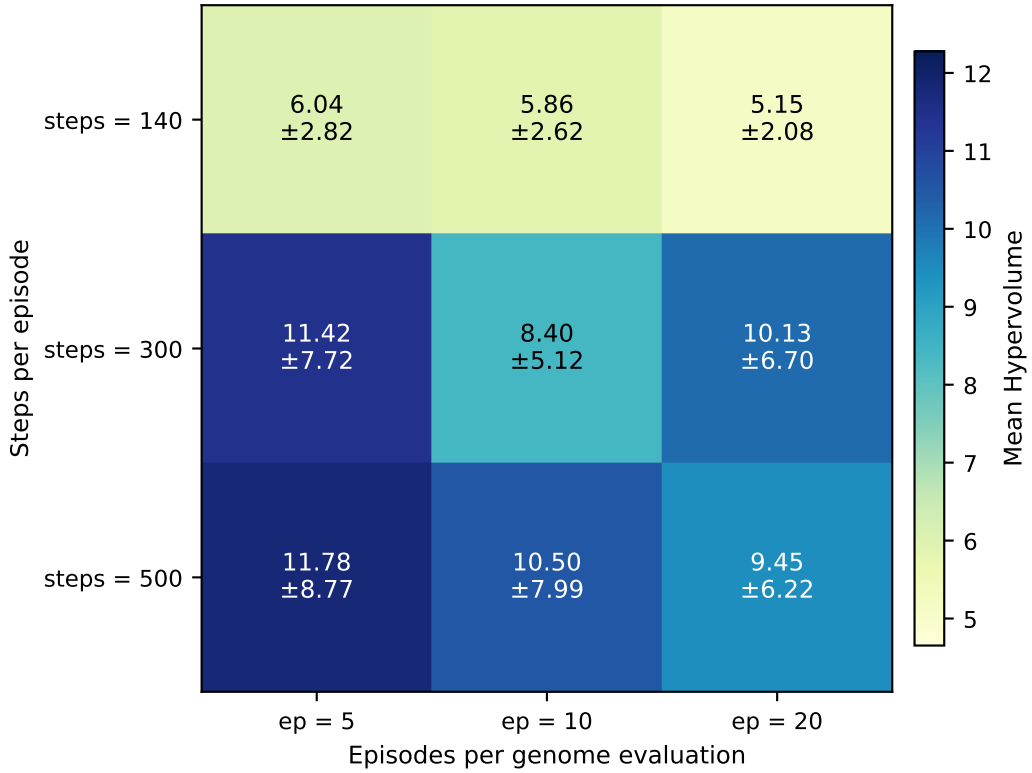

Figure 4: Scale sensitivity heatmap: mean HV by simulation horizon (steps $\times$ episodes/genome). Each cell is $n = 10$ independent NSGA-II runs. HV increases monotonically with steps; episode count has weaker secondary effect.

6 CONCLUSION

HEAS addresses the coupling code problem through four architectural contributions—Layer/Stream/Arena composition, a uniform metric contract, native NSGA-II integration, and tournament evaluation primitives. Across five evaluation questions, results confirm the framework hypothesis: 97% coupling code reduction (160 vs. 5 LOC), reproducible multi-run behavior at two simulation scales, stable tournament rankings under realistic noise ($\tau > 0.94$), and algorithm flexibility when landscape structure favors simpler optimizers. Three case studies—including a zero-coupling-code port of the published Mesa Wolf-Sheep model—and a 32-scenario OOD champion stress test ($p < 10^{-9}$) provide broad empirical coverage.

Limitations. For lightweight ODE simulations (<0.01 s/episode), process startup overhead ($\approx$9 s) dominates; use `n_jobs=1`. HEAS is designed for ODE-style and tabular agent simulations; spatially-explicit ABMs with rich movement and neighborhood interaction remain Mesa's domain. The metric contract is scalar-focused (`dict[str, float]`); workflows needing agent-level distributions require an additional aggregation layer.

Future work. Priorities are a `MesaLayerAdapter`, scalable parallel backends (Ray/Dask), and automated scenario generation.

For double-blind review, repository access details are omitted. An anonymized artifact package containing all scripts under `experiments/` is prepared for reviewers.

A REPORTED MEANS AND SOURCES

Source IDs (all paths under `experiments/results/`):

Table 2: Principal means reported in this paper with source file IDs (S1–S8).

Quantity	Value	Source
Coupling LOC reduction	97% (160 vs. 5)	S1
Ecological HV mean (small scale, $n=30$)	7.665 ± 3.518	S2
Enterprise HV mean (small scale, $n=2$)	3762.99	S2b
Noise τ at $\sigma=10$	0.944	S3
Noise τ at $\sigma=50$	0.744	S3
Simple ablation HV mean	19.66	S4
NSGA-II ablation HV mean	9.99	S4
Random ablation HV mean	7.61	S4
Large-scale NSGA-II HV (1,000 steps, $n=20$)	16.66 ± 10.84	S5
Large-scale Random HV (1,000 steps, $n=20$)	6.00 ± 0.34	S5
NSGA-II vs. Random Cohen's d (large scale)	1.39	S5
16-scenario OOD wins / Cohen's d	16/16 / 0.024	S6a
32-scenario OOD wins / Cohen's d	32/32 / 0.17	S6b
32-scenario mean Δ biomass	+25.0 (+3.96%)	S6b
Wolf-Sheep NSGA-II vs. Random p-value	7.99×10^{-6}	S7
Enterprise HV mean (large scale, $n=20$)	5.565×10^{6}	S8

S1 `mesa_vs_heas/loc_comparison.json`
S2 `eco_stats/pop20_ngen10_trait/summary.json`
S2b `ent_stats/summary.json`
S3 `tournament_stress/noise_stability.json`
S4 `baseline_comparison/ablation.json`
S5 `large_scale_comparison/p1_summary.json`
S6a `baseline_comparison/champion_vs_ref.json`
S6b `large_scale_comparison/p3_summary.json`
S7 `wolf_sheep_study/algo_comparison.json`
S8 `large_scale_comparison/p2_summary.json`

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AUTHOR BIOGRAPHIES

RUIYU ZHANG is affiliated with the Department of Politics and Public Administration at The University of Hong Kong, Hong Kong SAR, China. Her e-mail address is Ruiyuzh@connect.hku.hk.

LIN NIE is affiliated with the Department of Applied Social Sciences at The Hong Kong Polytechnic University, Hong Kong SAR, China. His e-mail address is lin-apss.nie@polyu.edu.hk.

XIN ZHAO is affiliated with the Department of Applied Social Sciences at The Hong Kong Polytechnic University, Hong Kong SAR, China. His e-mail address is xinnn.zhao@connect.polyu.hk.